INDEX · 144 QUESTIONS · 153 FILES

C++ Practice Programs

Every file in the zip, grouped by topic

Q1.cpp → Q144.cpp

WHAT'S IN THE ZIP

153 files, numbered by question

144

Questions

Q1 to Q144, one program each

150

C++ source files

Six questions have two versions

3

Other files

a.c, a.cpp and a.exe

How the files are named

`Q95i, Q95ii = two versions of question 95 · all are .cpp`

Topic map

Questions	Topic	Files
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Q1 TO Q13 · 13 FILES

Basics and arithmetic

File	What it does
Q1	Hello World
Q2	Print numbers and expressions
Q3	Assign and print a variable
Q4	Integer arithmetic on 5 and 2
Q5	Float arithmetic on 5 and 2
Q6	Increment and decrement
Q7	Area of a circle

File	What it does
Q8	Simple interest from p, r, t
Q9	Sum of two floats
Q10	Remainder with %
Q11	Cast int to float
Q12	Casting, second copy
Q13	5/2 versus (float) 5/2

Q14 TO Q30 · 17 FILES

Conditions and calculators

File	What it does
Q14	Even or odd
Q15	Positive and divisible by 5
Q16	Absolute value
Q17	Profit or loss from sp and cp
Q18	Area and perimeter of a rectangle
Q19	Is it a three-digit number
Q20	Divisible by both 5 and 3
Q21	Valid triangle from 3 sides
Q22	Greatest of three numbers

File	What it does
Q23	Divisible by 5 or 3, not both
Q24	Grade from percentage
Q25	Where a point (x, y) lies
Q26	Even or odd with ternary
Q27	Calculator using switch
Q28	Calculator using functions
Q29	Calculator using if-else ladder
Q30	Chained assignment trick

Q31 TO Q46 · 16 FILES

Loops, tables, AP and GP

File	What it does
Q31	Hello World 10 times
Q32	Hello World n times
Q33	Numbers 1 to 100
Q34	Even numbers to 100
Q35	Odd numbers to 100
Q36	Table of 19
Q37	Table of n, negatives too
Q38	AP: terms and sum

File	What it does
Q39	AP: terms and sum, version 2
Q40	AP: terms and sum, version 3
Q41	GP: terms and sum
Q42	Highest proper factor
Q43	Composite number check
Q44	Composite check with a flag
Q45	Odd numbers using continue
Q46	while + continue (never ends)

Q47 TO Q54 · 8 FILES

Digits, factorials and ASCII

File	What it does
Q47	Count the digits of a number
Q48	Sum of digits
Q49	Product of digits
Q50	Reverse a number

File	What it does
Q51	Show original and reversed number
Q52	Factorial of n
Q53	Factorials from 1 to n
Q54	ASCII values up to Z

Q55 TO Q73 · 19 FILES

Star, number and letter patterns

File	What it prints
Q55	3 by 5 rectangle of stars
Q56	4 by 4 square of stars
Q57	Square of rows 1234
Q58	Stars, m rows by n columns
Q59	Rows of 1 2 3 … n
Q60	Row number repeated across
Q61	Rows of A B C D …
Q62	One letter per row
Q63	Rows of a b c d …
Q64	One lowercase letter per row

File	What it prints
Q65	Right triangle of stars
Q66	Triangle of letters
Q67	Triangle of numbers
Q68	Alternating letter and number rows
Q69	Inverted star triangle (won't compile)
Q70	Inverted triangle of numbers
Q71	Inverted triangle of letters
Q72	Inverted triangle, repeated digits
Q73	Triangle counting down from n

Q74 TO Q90 · 17 FILES

Pyramids, diamonds and shapes

File	What it prints
Q74	Triangle of odd numbers
Q75	Floyd's triangle (1, 2, 3 ...)
Q76	Triangle of alternating 0 and 1
Q77	Plus sign of stars
Q78	Hollow box of stars
Q79	X-like star pattern
Q80	Right-aligned star triangle
Q81	Star diamond (86 lines)
Q82	Right-aligned number triangle

File	What it prints
Q83	Right-aligned letter triangle
Q84	Slanted block of stars
Q85	Star pyramid
Q86	Number pyramid
Q87	Letter pyramid
Q88	Empty (note: see Q81)
Q89	Number diamond
Q90	Letter diamond

Q91 TO Q100 · 12 FILES

Functions, nCr, swap and pointers

File	What it does
Q91	A function calling another
Q92	return ends a function early
Q93	Function with two parameters
Q94	Tour of cmath functions
Q95i	nCr using a factorial function
Q95ii	nCr, second version

File	What it does
Q96i	Pascal's triangle
Q96ii	Pascal's triangle, centred
Q97	Swap two numbers by reference
Q98	Pointer values and addresses
Q99	Swap two entered numbers
Q100	Greater of two numbers

Q101 TO Q108 · 8 FILES

Recursion

File	What it does
Q101	Print 1 to n and n to 1
Q102	Recursive sum with accumulator
Q103	Recursive sum of 1 to n
Q104	Recursive factorial

File	What it does
Q105	Power using pow()
Q106	Recursive Fibonacci
Q107	Fibonacci using a loop
Q108	Own power function, myPow

Q109 TO Q122 · 17 FILES

Arrays

File	What it does
Q109	Print, then overwrite 4 elements
Q110	Read 10 numbers into an array
Q111	Array changed inside a function
Q112i	Sum of array elements
Q112ii	Sum, size entered by user
Q113i	Product of array elements
Q113ii	Product, size entered by user
Q114	Maximum in an array
Q114ii	Minimum in an array

File	What it does
Q115	Array of squares up to n
Q116	Change elements by index parity
Q117	Count elements greater than x
Q118	Even-index sum minus odd-index sum
Q119	Second largest element
Q120	Reverse into a new array
Q121	Reverse in place by swapping
Q122	Palindrome array check

Q123 TO Q131 · 10 FILES

2D arrays and matrices

File	What it does
Q123	Print a 2D array column by column
Q124	Ways to initialise a 2D array
Q125	Student marks table
Q126	5 by 5 matrix filled with 10
Q127	Add two matrices

File	What it does
Q128i	Largest element in a matrix
Q128ii	Smallest element in a matrix
Q129	Product of all elements
Q130	Sum of the inner 2 by 2 block
Q131	Read and print a matrix

Q132 TO Q144 · 13 FILES

Strings

File	What it does
Q132	Create and print a string
Q133	Read one word with cin
Q134	Read a full line with getline
Q135	Count the vowels
Q136	Replace even positions with a
Q137	push_back, pop_back, append, clear
Q138	Join strings with +

File	What it does
Q139	Reverse a string
Q140	Reverse the first half
Q141	Reverse a slice of the string
Q142	Number to string, then append
Q143	Count digits with to_string
Q144	String to number with stoi

OUTSIDE THE NUMBERED QUESTIONS · 3 FILES

Scratch files

File	What it is
a.c	Sum of two numbers, repeated by calling main() again (C, not C++)
a.cpp	Scratch file: old pattern code in comments, one live program at the end
a.exe	A compiled Windows program